

Semitotal Domination in Graphs

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To our treasured friend Teresa, with thanks

ABSTRACT. In this paper we introduce a parameter that is squeezed between arguably the two most important domination parameters, namely the domination number and the total domination number. We define a set S of vertices in a graph G with no isolated vertices to be a *semitotal dominating set* of G if it is a dominating set of G and every vertex in S is within distance 2 of another vertex of S . The semitotal domination number, denoted by $\gamma_{t2}(G)$, is the minimum cardinality of a semitotal dominating set of G . We show that if G is a connected graph on $n \geq 4$ vertices, then $\gamma_{t2}(G) \leq n/2$, and we characterize the trees and graphs of minimum degree 2 achieving this bound.

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1 Introduction

In this paper we explore a parameter that is a strengthening of domination but a relaxation of both total domination and weakly connected domination. These notions are defined as follows. A *dominating set* of a graph G is a set S of vertices of G such that every vertex in $V(G) \setminus S$ is adjacent to at least one vertex in S . The *domination number* of G , denoted

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by $\gamma(G)$, is the minimum cardinality of a dominating set. A *weakly connected dominating set*, abbreviated WCD-set, is a dominating set S with the property that the graph $\langle S \rangle_w$ is connected, where $\langle S \rangle_w$ has vertex set S and its neighbors, and edge set those edges incident with S . Thus, for any WCD-set of cardinality at least 2, every vertex in S must be distance at most two from a vertex of S . The *weakly connected domination number* of G , denoted by $\gamma_w(G)$, is the minimum cardinality of a WCD-set. A *total dominating set*, abbreviated TD-set, of G is a set S of vertices of G such that every vertex in $V(G)$ is adjacent to at least one vertex in S . The *total domination number* of G , denoted by $\gamma_t(G)$, is the minimum cardinality of a TD-set.

We define here a set S of vertices in a graph G with no isolated vertices to be a *semitotal dominating set*, abbreviated semi-TD-set, of G if it is a dominating set of G and every vertex in S is within distance 2 of another vertex of S . The semitotal domination number, denoted by $\gamma_{t2}(G)$, is the minimum cardinality of a semi-TD-set. A semi-TD-set of G of cardinality $\gamma_{t2}(G)$ we call a $\gamma_{t2}(G)$ -set. For the rest of this paper, we assume that graphs have no isolated vertices.

Every semi-TD-set is also a dominating set. Similarly, every TD-set, and every WCD-set of cardinality at least 2, is a semi-TD-set. Hence the semitotal domination number is bounded below by the domination number and above by both the total domination number and the maximum of 2 and the weakly connected domination number:

Observation 1 *If G is a graph with no isolated vertices, then*

$$\gamma(G) \leq \gamma_{t2}(G) \leq \begin{cases} \gamma_t(G) \\ \max(\gamma_w(G), 2). \end{cases}$$

We proceed as follows. In Section 2 we provide bounds on the parameter. In Section 3 we show that if G is a connected graph on $n \geq 4$ vertices, then $\gamma_{t2}(G) \leq n/2$, and we characterize the trees and graphs of minimum degree 2 achieving this bound. In Section 4 we provide more bounds, and in Section 5 we discuss the computational complexity of the parameter.

1.1 Terminology and Notation

For notation and graph theory terminology not defined herein, we refer the reader to [5]. Let G be a graph with vertex set $V = V(G)$ of order $n = |V|$ and edge set E . We denote the *degree* of vertex v in G by $d_G(v)$, we

denote the maximum (minimum) degree among the vertices of G by $\Delta(G)$ ($\delta(G)$, respectively), and we call a vertex of degree one a *leaf*. The *open neighborhood* of v is the set $N_G(v) = \{u \in V \mid uv \in E\}$ and the *closed neighborhood* of v is $N_G[v] = \{v\} \cup N_G(v)$. For a set $S \subseteq V$, its *open neighborhood* is the set $N_G(S) = \bigcup_{v \in S} N_G(v)$, while its *closed neighborhood* is the set $N_G[S] = S \cup N_G(S)$. If the graph G is clear from the context, we omit the subscript G . For subsets $X, Y \subseteq V$, we denote the set of edges that join a vertex of X and a vertex of Y by $[X, Y]$.

2 Preliminary Results

The semitotal domination number of a path and a cycle is straightforward to compute.

Lemma 2 *For $n \geq 3$, $\gamma_{t2}(P_n) = \gamma_{t2}(C_n) = \lceil 2n/5 \rceil$.*

A semi-TD-set of the path/cycle when n is a multiple of 5 can be formed by numbering the vertices in order and taking those equivalent to 2 and 4 modulo 5. We omit the details. Straight from the definition, we see that $\gamma_{t2}(G) \geq 2$. This is achieved for example by stars. The lower bound for paths and cycles generalizes to the following result:

Lemma 3 *If G has n vertices and maximum degree Δ , then*

$$\gamma_{t2}(G) \geq \frac{2n}{2\Delta + 1}.$$

Proof. Let $G = (V, E)$ and let S be any semi-TD-set in G . For each vertex $u \in V$, let $f: V \rightarrow [0, 1]$ be defined by $f(u) = 1/k$, where $k = |N[u] \cap S|$. Thus if u is dominated by k vertices of S , then $f(u) = 1/k$. Let $v \in S$. If every vertex in $N[v]$ is dominated by v but by no other vertex of S , then v would be at distance at least 3 of another vertex of S , a contradiction. Hence there is at least one vertex, v' , in $N[v]$ that is adjacent to a vertex of S different from v . We note that $f(v') \leq 1/2$. The amount of domination done by v is therefore

$$\sum_{u \in N[v]} f(u) \leq f(v') + \sum_{u \in N[v] \setminus \{v'\}} f(u) \leq d_G(v) + \frac{1}{2} \leq \Delta + \frac{1}{2},$$

and so v effectively dominates at most $\Delta + 1/2$ vertices. The total amount of domination done by S is therefore at most $|S|(\Delta + 1/2)$. Since S is

a dominating set in G , we therefore have that $n \leq |S|(\Delta + 1/2)$, and so $\gamma_{t2}(G) = |S| \geq 2n/(2\Delta + 1)$. $\square$

The relationship between domination parameters is further exhibited by:

Lemma 4 *If G is connected and nontrivial, then*

- (a) $\gamma_{t2}(G) \leq 2\gamma(G) - 1$;
 - (b) $\gamma_t(G) \leq 2\gamma_{t2}(G) - 1$;
- and these bounds are sharp.*

Proof. (a) This result follows from Dunbar et al. [3], since they show that $\gamma_w(G) \leq 2\gamma(G) - 1$. Their example of equality carries over: Take k copies of the path P_5 and one P_2 , and identify one end of each path as the same vertex. The bound in the above lemma is also achieved by taking two disjoint isomorphic stars and identifying one leaf of each. In general, one can take multiple such trees and add edges between the leaves to achieve connectivity.

(b) Consider a minimum semi-TD-set S . By definition, there must be two vertices in S that are within distance 2 from each other, say s_1 and s_2 . Thus we obtain a total dominating set from S by taking the common neighbor of s_1 and s_2 (if it exists) and one neighbor for every other vertex in S . For an example of equality, take the star and subdivide all edges twice, except for one edge. $\square$

3 Graphs with Semitotal Domination Number Half their Order

For an upper bound, Dunbar et al. [3] observed that $\gamma_w(G) \leq n/2$ for a graph of n vertices. This follows from considering a spanning tree and noting that in any tree, each color class is a WCD-set. Thus by Observation 1 we obtain the following result.

Lemma 5 *For a graph G with every component of order $n \geq 4$, we have $\gamma_{t2}(G) \leq n/2$.*

3.1 Trees

In order to characterize the trees with semitotal domination number exactly one-half their order, we define a family $\mathcal{T}$ of trees as follows. Let H be a nontrivial tree and for each vertex v of H , add either a P_2 or a P_4 and

identify v with one end of the path. Let T denote the resulting tree and let $\mathcal{T}$ be the family of all such trees T . We call the tree H used to construct the tree T the *underlying tree* of T and let Y denote the vertices of P_4 -additions at distance two from their v . A tree in the family $\mathcal{T}$ is illustrated in Figure 1, where the vertices of the underlying tree are black and the vertices of Y are gray.

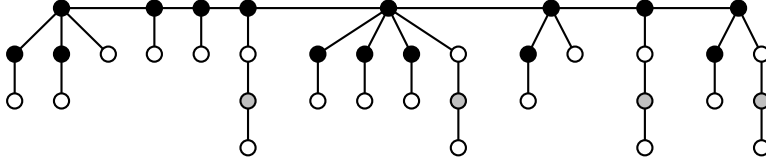


Figure 1: A tree in the family $\mathcal{T}$.

We observe that if $T \in \mathcal{T}$ has order n , then $\gamma_{t2}(T) = n/2$. To see this, note that every leaf has to be dominated. So we have to take at least one vertex from every P_2 -addition. Further, the leaf in each P_4 -addition has to have two vertices within distance 3 of it, so we need to take at least two vertices from every P_4 -addition. Conversely, the vertices of the underlying tree along with the set Y form a semi-TD-set.

We are now in a position to prove the following result.

Theorem 6 *Let T be a tree of order $n \geq 4$. Then, $\gamma_{t2}(T) = n/2$ if and only if $T \in \mathcal{T}$ or $T = K_{1,3}$.*

Proof. We proceed by induction on the order $n \geq 4$ of a tree T . If $n = 4$, then $T = P_4 \in \mathcal{T}$ or $T = K_{1,3}$ and in both cases $\gamma_{t2}(T) = 2 = n/2$. If $n = 5$, then $\gamma_{t2}(T) = 2 < n/2$. If $n = 6$, then $\gamma_{t2}(T) = 3$ if and only if $T = P_6$ or T is subdivided $K_{1,3}$; both are in $\mathcal{T}$. This establishes the base cases. Suppose that $n \geq 7$ and that for all trees T' of order n' , where $4 \leq n' < n$, we have $\gamma_{t2}(T') = n'/2$ if and only if $T' \in \mathcal{T}$ or $T' = K_{1,3}$. Let T be a tree of order n .

Let P be a longest path in T and let r and u be the two ends of P . We root T at the leaf r . Let v be the parent of u in T , and let w be the parent of v .

Suppose that v has another leaf-neighbor u' and consider the tree $T' = T - u$. By Lemma 5, there is a semi-TD-set S' of T' of cardinality at most $(n-1)/2$. If $v \notin S'$, then in order to dominate the leaf u' in T' we note that $u' \in S'$. But then replacing u' in S' with v produces a new semi-TD-set of T' . Hence we may assume that $v \in S'$. Thus, S' is a semi-TD-set of T and $\gamma_{t2}(T) \leq |S'| < n/2$. Thus we may assume that $d_T(v) = 2$. Let $T' = T - \{u, v\}$ have order n' and note that $n' = n - 2 \geq 5$.

Let S' be a semi-TD-set of T' . In order to dominate the vertex w , the set S' contains the vertex w or a neighbor of w (or both), and so the vertex v is within distance 2 in T from a vertex of S' . Hence $S \cup \{v\}$ is a semi-TD-set of T , implying that $\gamma_{t2}(T) \leq \gamma_{t2}(T') + 1 \leq n'/2 + 1 = n/2$. If $\gamma_{t2}(G') < n'/2$, then $\gamma_{t2}(T) < n/2$. Hence we may assume that $\gamma_{t2}(T') = n'/2$.

By the inductive hypothesis, $T' \in \mathcal{T}$. Let H' be the underlying tree of T' and consider the position of the vertex w in T' . If w is in H' , then T can be constructed as a member of $\mathcal{T}$ in the same way as T' except that we first add a vertex v to H' to form the underlying tree H and in the construction add a P_2 -addition at v as well. If w is a leaf-neighbor of some vertex x in H' , then T can be constructed as a member of $\mathcal{T}$ in the same way as T' , except that instead of using a P_2 -addition at x we use a P_4 -addition.

It remains to consider the case that vertex w is not in H' but on a P_4 -addition. Let x be the vertex of H' on this P_4 -addition, and let y be the vertex of Y . Note that x and y are at distance two apart in T . Let $S' = V(H') \cup Y(T')$. As observed earlier, the set S' is a semi-TD-set of minimum cardinality $\gamma_{t2}(T')$ in T' . Suppose first that H' has at least three vertices. Then every vertex on H' is within distance two of some vertex of H' other than x , implying that $S' \setminus \{x\}$ is a dominating set in T' and every vertex of $S' \setminus \{x\}$ except y is within distance two from some other vertex of $S' \setminus \{x\}$. Further we note that either $w = y$ or the vertex w is a neighbor of y , and so the vertex v is within distance two from y in T . This implies that the set $(S' \setminus \{x\}) \cup \{v\}$ is a semi-TD-set of T . Therefore in this case, $\gamma_{t2}(T) \leq |S'| < n/2$. Hence we may assume that H' has two vertices, and so T' is either P_6 or P_8 .

Assume $T' = P_6$. If w is a leaf of T' , then $T = P_8$. If w has a leaf-neighbor, then T is obtained from a path P_3 by using a P_2 -addition at the central vertex and one of its leaf neighbors and by using a P_4 -addition at the remaining leaf (as illustrated in Figure 2(a)). If w is a central vertex of T' , then T is obtained from a path P_3 by using a P_4 -addition at the central vertex and a P_2 -addition at the two leaves (as illustrated in Figure 2(b)). Assume $T' = P_8$. If w is not one of the two central vertices of T' , then $\gamma_{t2}(T) \leq 4 < n/2$. If w is a central vertex of T' , then T is obtained from a path P_3 by using a P_4 -addition at the central vertex and one of its leaf neighbors and by using a P_2 -addition at the remaining leaf (as illustrated in Figure 2(c)). In all cases $T \in \mathcal{T}$. $\square$

3.2 Minimum Degree Two

A natural question is what is the maximum semitotal domination number in large connected graphs of minimum degree at least 2. However, this

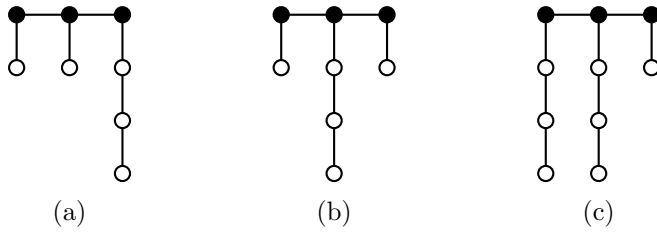


Figure 2: Trees in the family $\mathcal{T}$.

turns out to be straight-forward, as one can construct such graphs with semitotal domination number half their order.

Define the generalized *corona* of G with H as follows. For every vertex in G , form a copy of H and identify one vertex in the copy of H with the corresponding vertex in G . We denote this as $\text{cor}_H(G)$. As long as H is vertex-transitive, this notation is well defined. It is not hard to show that any generalized corona with 4-cycle has semitotal domination number half its order. See Figure 3.

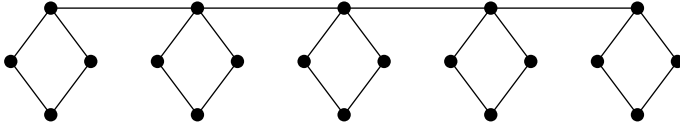


Figure 3: The generalized corona $\text{cor}_{C_4}(P_5)$

In fact, we can determine all graphs of minimum degree 2 achieving the upper bound.

Theorem 7 *Let G be a connected graph of order $n \geq 4$ with minimum degree at least 2. Then $\gamma_{t2}(G) = n/2$ if and only if G is C_6 , C_8 , a spanning subgraph of K_4 , or a generalized corona with 4-cycle.*

In order to prove Theorem 7, we first present two preliminary results. The following observation is easily checked:

Observation 8 *For any cycle other than C_3 , C_5 , and C_7 , one can choose a semi-TD-set of at most half the vertices containing any two specified vertices.*

We shall also need the following key lemma.

Lemma 9 *Let G be a (not necessarily connected) graph of order n with every component of order at least 4 that is edge-minimal with respect to having minimum degree at least 2. If $\gamma_{t2}(G) = n/2$, then G is 2-regular.*

PROOF. We prove by induction that if the graph G has $\Delta(G) > 2$, then $\gamma_{t2}(G) < n/2$. If $n = 4$, then the edge-minimality of G implies that $G = C_4$. This establishes the base case. Suppose then that $n \geq 5$ and that the result holds for all graphs of order less than n every component of which has order at least 4 and that are edge-minimal with respect to having minimum degree at least 2. Let G be a graph of order n with every component of order at least 4 that is edge-minimal with respect to having minimum degree at least 2 and suppose that $\Delta(G) > 2$. If G is disconnected, then we can apply the inductive hypothesis to the components that are not cycles to yield the desired result; so we may assume that G is connected. If n is odd, then by Lemma 5, $\gamma_{t2}(G) < n/2$ and we are done. Hence we may assume that n is even. In particular, $n \geq 6$.

We proceed further with the following claim.

Claim A *We can assume that any two adjacent vertices of degree 2 either lie in a 3-cycle, or lie in a 4-cycle with another vertex of degree 2.*

Proof of Claim A. Suppose vertices x and y of degree 2 are adjacent. If they have a common neighbor, then we have the desired 3-cycle. So assume they do not have a common neighbor. Let a be x 's other neighbor, and b be y 's other neighbor. If a is adjacent to b , then at least one of a or b has degree 2, and we have the desired 4-cycle. So, suppose that a is not adjacent to b .

There are two cases. Assume first that at least one of a or b has degree 2. In this case let G' be the graph of order n' obtained from G by deleting x and y and adding the edge ab . By construction, the graph G' is a connected graph of order $n' = n - 2 \geq 4$ with $\Delta(G') > 2$ that is edge-minimal with respect to having minimum degree at least 2. Applying the inductive hypothesis to G' , we have that $\gamma_{t2}(G') < n'/2$. Let S' be a $\gamma_{t2}(G')$ -set. If $a \in S'$, let $S = S' \cup \{y\}$. If $a \notin S'$, let $S = S' \cup \{x\}$. In both cases, the set S is a semi-TD-set of G , implying that $\gamma_{t2}(G) \leq |S| = |S'| + 1 < n'/2 + 1 = n/2$ and therefore that the desired result of the lemma follows.

The second case is that both a and b have degree at least 3. In this case we consider the graph $G'' = G - \{x, y\}$. Let G'' have order n'' , and so $n'' = n - 2 \geq 4$. Further let G_a and G_b denote the component of G'' containing a and b , respectively. Then either G'' is connected, in which case $G'' = G_a = G_b$, or G'' is disconnected, in which case $G'' = G_a \cup G_b$. If both

G_a and G_b are 3-cycles, then $n = 8$ and $\{a, b, x\}$ is a semi-TD-set of G'' , implying that $\gamma_{t2}(G) \leq 3 < n/2$. If exactly one of G_a and G_b is a 3-cycle, say G_a , then by Lemma 5 we have that $\gamma_{t2}(G_b) \leq |V(G_b)|/2 = (n-5)/2$. Every $\gamma_{t2}(G_b)$ -set can be extended to a semi-TD-set of G by adding to it the vertices a and y , implying that $\gamma_{t2}(G) \leq \gamma_{t2}(G_b) + 2 < n/2$. Hence we may assume that both G_a and G_b have order at least 4. Thus every component of G'' has order at least 4 and G'' is edge-minimal with respect to having minimum degree at least 2.

Suppose that $\Delta(G'') > 2$. Applying the inductive hypothesis to G'' , we have that $\gamma_{t2}(G') < n''/2$. Every $\gamma_{t2}(G')$ -set can be extended to a semi-TD-set of G by adding to it either x or y , implying that $\gamma_{t2}(G) \leq \gamma_{t2}(G') + 1 < n''/2 + 1 = n/2$. Hence we may assume that G'' is 2-regular, for otherwise the desired result of the lemma follows. Since G is connected, it follows that G'' is either a single cycle or the union of two cycles. By our earlier assumptions, none of the cycles is a 3-cycle. Suppose none of the cycles is a 5- or 7-cycle. Then by Observation 8, we can choose a semi-TD-set S of half the vertices of G'' containing both a and b ; then, S is also a semi-TD-set of G , implying once again that $\gamma_{t2}(G) \leq |S| = n''/2 < n/2$. Hence we may assume that G'' contains a 5- or a 7-cycle. Recall that n is even, and so n'' is even. Thus, G'' is the disjoint union of two odd cycles, at least one of which is a 5- or a 7-cycle. By Lemma 5, $\gamma_{t2}(G_x) < |V(G_x)|/2$ for $x \in \{a, b\}$. Let S_a and S_b be a $\gamma_{t2}(G_a)$ - and $\gamma_{t2}(G_b)$ -set, respectively. Since both G_a and G_b are cycles, by symmetry we can choose the set S_a to contain a and the set S_b to contain b . But then $S_a \cup S_b$ is a semi-TD-set of G , implying that $\gamma_{t2}(G) \leq |S_a| + |S_b| < n''/2 < n/2$. $\square$

We now return to the proof of Lemma 9. Recall that $\Delta(G) > 2$ and $n \geq 6$ is even. By Claim A, any two adjacent vertices of degree 2 either lie in a 3-cycle, or lie in a 4-cycle with another vertex of degree 2. Let $\mathcal{L}$ denote the set of vertices in G of degree at least 3. By minimality, the set $\mathcal{L}$ is an independent set. Since $\Delta(G) > 2$, we note that $\mathcal{L}$ is not empty. Let $Y = N(\mathcal{L})$ and let X be the vertices not dominated by $\mathcal{L}$. We note that every vertex in $X \cup Y$ has degree 2 in G . Further both neighbors of a vertex in X have degree 2 in G . By the connectivity of G and the fact that $n \geq 6$, Claim A implies that each vertex of X lies in a 4-cycle with two vertices of Y and with one vertex of $\mathcal{L}$. In particular, X is an independent set and the set $\mathcal{L} \cup X$ is a dominating set of G . Further every vertex of X is at distance 2 from a vertex of $\mathcal{L}$.

If $|\mathcal{L}| = 1$ and $X = \emptyset$, then by Claim A, the graph G is obtained from $|Y|/2$ disjoint triangles by identifying one vertex from each copy into a common vertex. But this contradicts the fact that n is even. Hence, $|\mathcal{L}| \geq 2$ or $|\mathcal{L}| = 1$ and $|X| \geq 1$.

Suppose that $|L| \geq 2$. Let v be an arbitrary vertex of L and let u be a vertex of L at minimum distance from v in G . Since L is an independent set, we note that $d(u, v) \geq 2$. If $d(u, v) \geq 3$, then the internal vertices of a shortest u - v path in G are all of degree 2 and lie in neither a 3-cycle nor a 4-cycle, contradicting Claim A. Hence, $d(u, v) = 2$. Therefore if $|L| \geq 2$, then every vertex in L is within distance 2 of another vertex of L . As observed earlier, every vertex of X is at distance 2 from a vertex of L . Hence since $|L| \geq 2$ or $|L| = 1$ and $|X| \geq 1$, the dominating set $L \cup X$ is a semi-TD-set of G .

Finally, we claim that $L \cup X$ is less than half the vertices. Since every vertex in Y has degree 2, every vertex in Y is adjacent to at most two vertices in $L \cup X$, and so $|[Y, L \cup X]| \leq 2|Y|$. On the other hand, every vertex in L is adjacent to at least three vertices in Y , while every vertex in X is adjacent to exactly two vertices in Y , and so $|[Y, L \cup X]| \geq 3|L| + 2|X| > 2|L| + 2|X|$ since $|L| \geq 1$. Consequently, $|L| + |X| < |Y| = n - (|L| + |X|)$, or, equivalently, $|L| + |X| < n/2$, as claimed. Hence, $\gamma_{t2}(G) \leq |L| + |X| < n/2$. This completes the proof of Lemma 9. $\square$

We are now present a proof of Theorem 7. Recall its statement.

Theorem 7 *Let G be a connected graph of order $n \geq 4$ with minimum degree at least 2. Then $\gamma_{t2}(G) = n/2$ if and only if G is C_6 , C_8 , a spanning subgraph of K_4 , or a generalized corona with 4-cycle.*

PROOF. It is easily seen that the graphs listed have semitotal domination number equal to half their order. So to prove Theorem 7, it suffices to prove that these are the only such graphs. Let $G = (V, E)$ be a connected graph of order $n \geq 4$ with minimum degree at least 2 satisfying $\gamma_{t2}(G) = n/2$. If $n = 4$, then G is a spanning subgraph of K_4 and there is nothing to prove. Hence we may assume that $n \geq 6$. Let H be a spanning subgraph of G that is edge-minimal with respect to having minimum degree at least 2 and $\gamma_{t2}(H) = n/2$. By Lemmas 2 and 9, it follows that H is the disjoint union of 4-, 6-, and 8-cycles.

Assume first that H contains an 8-cycle C . Suppose that a vertex v of C is adjacent to a vertex v' outside C . Let S be a $\gamma_{t2}(H - V(C))$ -set. By symmetry, we may choose the set S to contain the vertex v' . But then adding to S the set of three vertices in C at even (positive) distance from v in C produces a semi-TD-set of G of cardinality $|S| + 3 = (n-8)/2 + 3 < n/2$, a contradiction. Hence, $V = V(C)$. If $G \neq C$, then since adding any edge to an 8-cycle reduces its semitotal domination number we have that $\gamma_{t2}(G) \leq 3 < n/2$, a contradiction. Hence, $G = C$.

Assume second that H contains a 6-cycle C . Suppose that a vertex v of C is adjacent to a vertex v' outside C . Let S be a $\gamma_{t2}(H - V(C))$ -set that

contains v' . Adding to the set S the two vertices in C at distance 2 from v in C produces a semi-TD-set of G of cardinality $|S| + 2 = (n - 6)/2 + 3 < n/2$, a contradiction. Hence, $V = V(C)$. If $G \neq C$, then since adding any edge to a 6-cycle reduces its semitotal domination number we have that $\gamma_{t2}(G) = 2 < n/2$, a contradiction. Hence, again in this case $G = C$.

Finally, assume that every cycle in H is a 4-cycle. Recall that $n \geq 6$, and so H is the disjoint union of at least two 4-cycles. We now consider an arbitrary 4-cycle C in H . Suppose that two vertices, u and v , of the cycle C are adjacent to vertices, u' and v' say, outside C . Possibly, $u' = v'$. Let S be a $\gamma_{t2}(H - V(C))$ -set. Since every component of $H - V(C)$ is a 4-cycle, we can choose the set S to contain both u' and v' . But then the set S can be extended to a semi-TD-set of G by adding to it a vertex in $V(C)$ different from u and v , implying that $\gamma_{t2}(G) \leq |S| + 1 = (n - 4)/2 + 1 < n/2$, a contradiction. Hence exactly one vertex of C is adjacent to a vertex outside C . We call such a vertex of the cycle C the link vertex of the cycle.

Let L denote the set of link vertices in H . Since G is connected, the subgraph induced by L is connected. Further since H is the disjoint union of at least two 4-cycles, and since the set L contains one vertex from each cycle, we note that $|L| \geq 2$. For each link vertex v , let v' denote a neighbor of v in H and let L' denote the resulting set of $|L|$ selected neighbors of the link vertices. Let $D = L \cup L'$.

As before, consider an arbitrary 4-cycle C in H with link vertex v . Suppose that there is a vertex u in C different from v of degree at least three. As observed earlier, the vertex u is only adjacent in G to vertices in $V(C)$. Hence, u has degree 3 in G and is adjacent to every other vertex of C in G . But then the set $(D \setminus \{v, v'\}) \cup \{u\}$ is a semi-TD-set of G , implying that $\gamma_{t2}(G) \leq |D| - 1 = n/2 - 1 < n/2$, a contradiction. Hence every vertex of G that is not a link vertex has degree 2 in G , implying that G is a generalized corona with 4-cycle. This completes the proof of Theorem 7. $\square$

4 More Bounds

For general δ , we have the same maximum semitotal domination number as in domination, and this is best possible since Alon [1] showed that the domination bound is essentially best possible.

Theorem 10 *For a connected graph G with order n and minimum degree δ ,*

$$\gamma_{t2}(G) \leq n(1 + o(1)) \frac{\ln(\delta + 1)}{\delta + 1}.$$

where by $o(1)$ we mean some quantity that tends to 0 as δ gets large.

PROOF. The proof is almost identical to the standard probabilistic proof of the upper bound for the domination number (e.g. see [2]). Construct a set D by choosing each vertex independently with probability $p = \ln(\delta+1)/(\delta+1)$. The probability that any particular vertex v is not dominated is $(1-p)^{\delta+1} \leq e^{-p \ln(\delta+1)} = 1/(\delta+1)$. Similarly, since the result is trivial if the graph is complete, we can assume that every vertex has at least one vertex at distance 2, and so the probability that any particular vertex w has no vertex of D at distance 1 or 2 from it is also at most $1/(\delta+1)$. If we add to D each vertex v that is not dominated, and one vertex x for each vertex w that has no vertex of D nearby, we obtain a set D' that is a semi-TD-set of G . And $|D'| \leq (2 + \ln(\delta+1))/(\delta+1)$. $\square$

For minimum degree 3 in particular, we conjecture:

Conjecture 1 *For all graphs G of order n and minimum degree at least 3, $\gamma_{t2}(G) \leq 2n/5$.*

Recall that a diamond is the graph obtained from a complete graph K_4 by removing an edge. If Conjecture 1 is true, then this bound is sharp. For example, take a graph H with no isolated vertex and for each vertex v of H , attach a diamond and join v to the two vertices of degree 2 in the diamond. Let the resulting graph G have order n . Then, $\gamma_{t2}(G) = 2n/5$. It is unclear what happens with cubic graphs. Computer evidence suggests that maybe for $n > 10$ all cubic graphs G of order n have $\gamma_{t2}(G) \leq 3n/8$.

We conclude this section with a result on the subdivision of a graph. Specifically we consider the subdivision where every edge is subdivided once. We use the terms “*original*” and “*subdivision*” vertices to refer to the vertices of the original graph and the added vertices respectively. For a graph G , let Q be a set of vertices that induces disjoint copies of P_3 in G and such that $G - Q$ has no isolated vertices (or has no vertex). Define $p_3(G)$ to be the number of copies of P_3 induced by the largest such Q .

Theorem 11 *If graph G has n vertices and H is the subdivision of G (with every edge subdivided once), then $\gamma_{t2}(H) = n - p_3(G)$.*

PROOF. We begin by forming a semi-TD-set of this size, proving this as an upper bound. Construct set S as follows. Let Q be a subset of the vertices of G such that the subgraph induced by Q is $p_3(G)$ copies of P_3 and $G - Q$ has no isolated vertices. Then, for every such P_3 , add to S the two subdivision vertices in H . Also, add to S all vertices of H corresponding

to vertices in $G - Q$. It is easily checked that S is a semi-TD-set of H . By construction $|Q| = n - p_3(G)$ and so, $\gamma_{t2}(H) \leq n - p_3(G)$.

We now show that every semi-TD-set is at least as large as this. Let T be a minimum semi-TD-set of H . Out of all such T , choose one that has *the minimum number of subdivision vertices*. If T contains two adjacent vertices, u and v , then exactly one is a subdivision vertex, say u ; if we chose both neighbors of u to be in T instead of u and v , we would still have a semitotal dominating set. So, T contains no adjacent vertices.

Let x be any subdivision vertex in T , if any. Then there is another vertex of T , say y , at distance 2 from x ; necessarily, y is a subdivision vertex too. That is, they must lie on a P_5 in H , say u, x, v, y, w . If u and w are adjacent in G , then the vertex between them in H must also be in T . But, in that case, we can choose u, v, w to be in T instead of these three. So, u and w are nonadjacent in G .

If u, v , or w has another neighbor in T , say z , then we can choose the other neighbor of z to be in T instead, thereby reducing the number of subdivision vertices. So, $N_H(\{u, v, w\}) \cap T = \{x, y\}$. It follows that the subdivision vertices in T occur in pairs, each on its own copy of what in G is a P_3 . Further, for every other edge incident with one of u, v or w , the subdivision vertex must be dominated by the other end of the edge. Thus if we consider the subgraph in G induced by the vertices incident with the edges whose subdivision vertices are in T , it follows that that subgraph is disjoint copies of P_3 . At the same time, all other vertices of G must be in T , since T dominates H . Thus $|T| = n - q$, where q is the number of copies of P_3 above. This means that $\gamma_{t2}(H) \geq n - p_3(G)$, and the result follows.

□

5 Complexity

It is not surprising that the problem of finding γ_{t2} is NP-complete. This can be shown by reduction from domination (known to be NP-complete; see for example [4]). Consider a graph G , and construct $\text{cor}_{P_3}(G)$. Identify all of the vertices on the free ends of the copies of P_3 . Call this vertex x and add another vertex adjacent only to it, and call this new graph G' . For any dominating set S of G , $S \cup \{x\}$ is a semitotal dominating set of G' . Also, if T' is a semi-TD-set of G' , then choose T to be every vertex in G which is in T' or adjacent to some vertex in $T' - V(G)$. Then T is a dominating set for G . Thus, $\gamma_{t2}(G') = \gamma(G) + 1$.

Conversely, there is a linear-time algorithm for trees, as expected. One approach is a Wimer-style algorithm (see [6]). We sketch the algorithm

and justification below. The idea in such an algorithm is to root the tree arbitrarily and then define for each vertex v , the set $\mathcal{F}(v)$ as the set of all restrictions of semi-TD-sets S to the subtree rooted at v . The key is that one can partition $\mathcal{F}(v)$ into a finite number of classes based on the state of v . Here these five states are:

- 1) v is undominated;
- 2) v is in S but no other vertex of S is within distance 2 of v ;
- 3) v is not in S , is dominated by S , but that neighbor has no other vertex in S within distance 2 of it;
- 4) v is in S and has another vertex in S within distance 2;
- 5) v is not in S but is dominated by S and that neighbor has another vertex in S within distance two of it.

The algorithm calculates, using a postorder traversal, at each vertex v the vector $(f_1, f_2, f_3, f_4, f_5)$, where f_i is the minimum cardinality of a member of $\mathcal{F}_i$. At the end, the semitotal domination number of the tree is the minimum of f_4 and f_5 at the root.

The formula for computing the f_i recursively is determined by the classes formed by combining two rooted trees, making one root the child of the other, with their sets. This is given in Figure 4, where we also provide an example of the smallest possible tree of type $\mathcal{F}_i$ (black vertices are in the set S).

6 Open Questions

We conclude with a few open questions. We have already posed the question of what is the maximum semitotal domination number for graphs with minimum degree 3 in general and for cubic graphs in particular. Another question of interest is what is the maximum number of edges in a graph for a given n and γ_{t2} .

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		Child					
							
		1	2	3	4	5	
Parent		1	–	3	–	5	1
		2	2	4	4	4	4
							
		3	–	5	–	5	3
							
		4	4	4	4	4	4
							
		5	–	5	–	5	5
Valid at end is: 4,5,							

Figure 4: Wimer table for semitotal domination

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